 NAME ______________ DATE ____________

1 Calcium homeostasis

COMP

You should be able to: Trace the response to falling plasma calcium through parathyroid hormone, calcitriol, and the actions on bone, kidney, and intestine, and state the role of calcitonin.

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4 · WHAT BREAKS IF THIS FAILS

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 NAME ____________ DATE ____________

2
COMP

Kidney functions

You should be able to: List the homeostatic functions of the kidney including water and electrolyte balance, acid base regulation, waste excretion, blood pressure control, and hormone production.

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3

COMP

Nephron structure and function

You should be able to: Match each nephron segment to its dominant transport function and distinguish a cortical from a juxtamedullary nephron by capability.

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4

COMP

Three renal processes

You should be able to: Define filtration, reabsorption, secretion, and excretion and write the equation that relates them for any solute.

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5
COMP

Filtration membrane and selectivity

You should be able to: Describe the three layers of the filtration membrane and predict which plasma components appear in the filtrate based on size and charge.

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6

COMP

Net filtration pressure and GFR

You should be able to: Calculate net filtration pressure from glomerular hydrostatic pressure, capsular pressure, and colloid osmotic pressure and predict the effect of each change on glomerular filtration rate.

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7
COMP

Regulation of GFR

You should be able to: Compare myogenic autoregulation, tubuloglomerular feedback, sympathetic control, and hormonal control and predict the effect of afferent and efferent arteriolar constriction on filtration rate.

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8

COMP

Renal clearance

You should be able to: Calculate the clearance of a substance and use inulin and creatinine clearance to estimate glomerular filtration rate.

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 NAME ______ DATE ______

9

COMP

Clearance and handling inference

You should be able to: Use a clearance value relative to glomerular filtration rate to determine whether a substance is net reabsorbed or net secreted.

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NAME ____________________ DATE ______________

10 COMP

Proximal tubule reabsorption

You should be able to: Explain how the sodium gradient drives reabsorption of glucose, amino acids, and water in the proximal tubule and state why reabsorption there is described as isosmotic.

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11

COMP

Transport maximum and renal threshold

You should be able to: Distinguish transport maximum from renal threshold and explain the appearance of glucose in urine when plasma glucose is elevated.

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12
COMP

Tubular secretion

You should be able to: State the purpose of tubular secretion and name substances secreted including hydrogen ion, potassium, and organic anions such as drugs.

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13
COMP

Countercurrent mechanism

You should be able to: Explain how the countercurrent multiplier of the loop of Henle and the countercurrent exchanger of the vasa recta build and preserve the medullary osmotic gradient.

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14
COMP

Antidiuretic hormone and urine concentration

You should be able to: Trace the response to dehydration from osmoreceptor firing through antidiuretic hormone release to aquaporin insertion and concentrated urine.

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15

COMP

Renin angiotensin aldosterone system

You should be able to: Trace the renin angiotensin aldosterone pathway from the stimulus for renin release to the effects of angiotensin II and aldosterone on vessels, tubules, and blood pressure.

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16

COMP

Natriuretic peptides

You should be able to: Explain how atrial natriuretic peptide opposes the renin angiotensin aldosterone system and state the stimulus for its release.

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17

COMP

Potassium handling

You should be able to: Explain how aldosterone and plasma potassium regulate potassium secretion in the collecting duct and predict the effect of a potassium wasting diuretic.

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18

COMP

Micturition

You should be able to: Trace the micturition reflex and explain how the internal and external sphincters allow voluntary control.

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19

COMP

Urinalysis

You should be able to: Perform a urinalysis measuring specific gravity, pH, glucose, protein, ketones, and blood and relate any abnormal finding to the renal process that failed.

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20

COMP

Renal function calculation

You should be able to: Calculate glomerular filtration rate, clearance, and filtered load from a data set and interpret the values in a clinical case.

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21
COMP

Volume and osmolarity disturbances

You should be able to: Classify a disturbance as volume depletion or overload and as hypoosmotic, isosmotic, or hyperosmotic and predict the resulting fluid shift between compartments.

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